

TEAM NAME: TEAM_GOJO

COLLEGE NAME: SVCE

PROBLEM STATEMENT: Man-made Disaster Management
System .



PROBLEM STATEMENT



! THREAT : 3,000,000+ open borewells and pits across India.

! BLIND SPOT : Zero surveillance exists between 6 PM and 6 AM.

😞 RESULT : This 12-hour blind spot is responsible for 80% of accidents, claiming 40+ young lives annually.



ECONOMIC INFEASIBILITY

- Proper sealing costs around ₹2.5–6 Lakhs per hole, leaving 98% of hazards open indefinitely



CURRENT SOLUTIONS FAILS

- Metal grates face 30-40% theft rates
- Temporary fences collapse in 70% of monsoons
- warning signs fade into irrelevance



CRITICAL WINDOW

- Survival rate drops below 20% after 24 hours of entrapment
- Rescue operations cost ₹10+ Lakhs and take 36-72 hours



SOLUTION

S.H.I.E.L.D. - SMART HAZARD INTELLIGENT EARLY LIFE DEFENSE



INNOVATION

A sub-₹1,000 autonomous safety unit that provides immediate interim security, bridging the critical vulnerability window between hazard identification and permanent municipal closure

**3-SECOND
INTERVENTION
PROTOCOL**



DETECT

Triple-redundant sensing (mmWave Radar + PIR + Ultrasonic) distinguishes a child from debris with (approx.) 90-95% accuracy



DETER

Active escalation — Yellow Strobe (**warning**) → 100dB Siren (**critical**) → Instant SMS Blast to Authorities



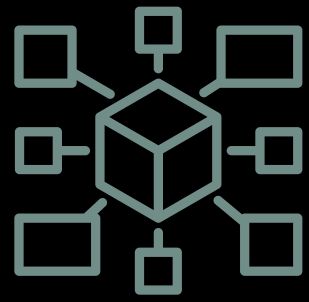
ENDURE

Solar-powered with 15-day battery backup, IP67 monsoon-proofing, and theft-proof concrete anchoring

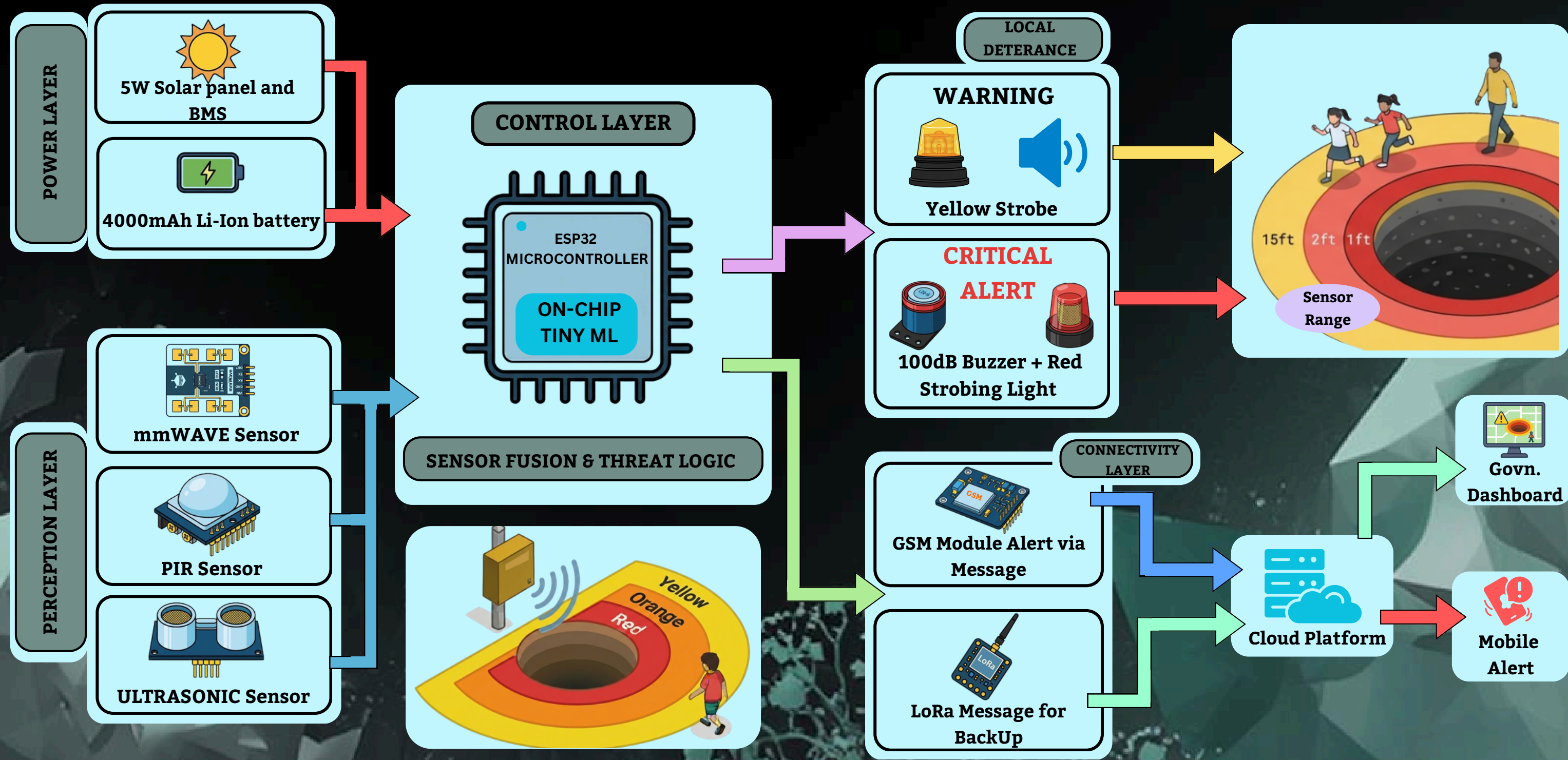


ECONOMIC EDGE

Achieve 8–15 times coverage density by replacing one CCTV with S.H.I.E.L.D. nodes— delivering active deterrence and zero-infrastructure deployment



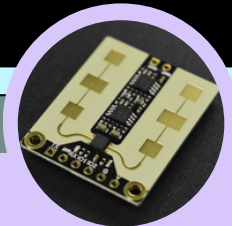
FLOW DIAGRAM





TECHNICAL REQUIREMENTS

PERCEPTION LAYER



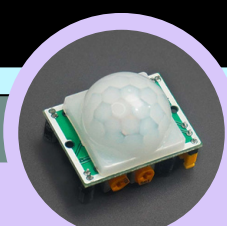
MMWAVE RADAR

Detects human micro-motion (breathing) within 5m through objects



ULTRASONIC SENSOR

Critical rim proximity (0-100cm)



PIR SENSOR

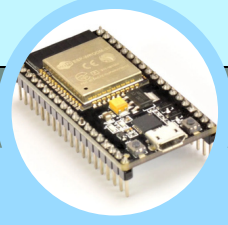
Validates body heat in the range



TILT SWITCH

Anti-theft and tamper detection

INTELLIGENCE LAYER



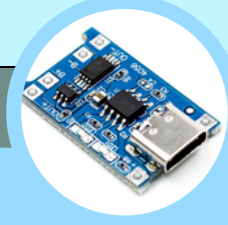
CONTROL UNIT

High-performance, dual-core microcontroller, with Tensor Flow Lite



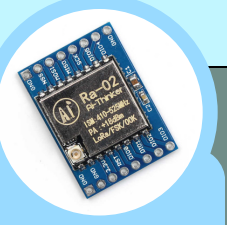
POWER MANAGEMENT

TP4056 BMS + deep-sleep firmware (90hr autonomy)



CONNECTIVITY

SIM800L GSM + SX1276 LoRaWAN (10km mesh)



ACTUATORS

100dB Piezo Siren and RGB Strobes

INFRASTRUCTURE



POWER

5W Solar Panel + 4000mAh 18650 Li-Ion (15-day backup)



IP67 CASE

Submersion-proof, -20°C to +65°C rated



MOUNTING

Ground Screws + Concrete Ballast impossible to steal

NOVELTY / USE CASE

PRECISION PARADOX

Delivering rugged casing with approx. 95% accuracy at a consumer-grade price point, shattering the cost-accuracy trade-off

ACTIVE VS PASSIVE DEFENSE

Unlike static fences or signs, S.H.I.E.L.D. creates an actively hostile environment for accidents with escalating 3-tier deterrence

THEFT-PROOF

Subterranean concrete ballast makes stealing the unit costs more to excavate than scrap value

CLOUD AUTONOMY

Edge AI eliminates internet dependency. LoRaWAN mesh enables village-wide coverage with a single GSM gateway

SCALABILITY SHIFT

Transforms safety from a capital expense (₹50k/sealing) to an operational expense (₹1K/guardian), enabling mass deployment.



BUSINESS MODEL / TARGET MARKET



Immediate Life-Saving

- Intervenes in the decisive 3-second window between approach and fall
- Targets 80% of accident scenarios (unmonitored) that traditional solutions miss entirely



Scalable Economics

- Secure an entire village for the price of one static CCTV camera (₹1k vs ₹15k)
- One month of human wages (₹12,000) funds 12 permanent guardians for remote hazard clusters



Instant & Unbreakable

- Zero-skill setup replaces 28-day concrete work
- Fully independent with 15-day solar autonomy
- Subterranean anchoring guarantees 5-year persistence in high-theft zones



Sustainable Safety

- 100dB alerts turns the community into first responders, outpacing rural ambulance arrival times
- Immediate protection that avoids 4-5 tons of CO2, during the 18-24 month closure wait

TEAM DETAILS

ROLE	NAME	STREAM /DEPT	YEAR	COLLEGE
Team Leader	NIHIL P RAYEN P	B.TECH/AD	III	SVCE
Team Member	MOHITH KUMAR G	B.TECH/AD	III	SVCE
Team Member	SANJAY D	B.TECH/AD	III	SVCE
Team Member	MUKESH S	B.TECH/AD	III	SVCE
Team Member	Vaishnavi Dinakaran	B.TECH/AD	III	SVCE
Team Member	MICHELLE JEFFRIN X	B.TECH/AD	III	SVCE